

Research Paper

Scalable, process-oriented beam lattices: Generation, characterization, and compensation for open cellular structures

Ian R. Woodward, Catherine A. Fromen^{*}

Department of Chemical and Biomolecular Engineering, University of Delaware, Newark, Delaware, United States

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ABSTRACT

Additively manufactured lattices are emerging as promising candidates for structural, thermal, chemical, and biological applications. However, achieving a satisfactory prototype or final part with this level of complexity requires synthesis of disparate knowledge from the distinctly digital and physical processing stages. This work proposes an integrated framework for processing self-supporting, open lattice structures that do not require supports and facilitate material removal in post-processing steps. We describe a minimal yet comprehensive design strategy for generating uniform lattice structures with conformal open lattice skins for an arbitrary unit cell configuration. Using continuous liquid interface production (CLIP™) on a Carbon M1, printability is evaluated for five unique bending-dominated lattice structures at unit cell length scales from 0.5 to 3.5 mm and strut diameters ranging from 0.11 to 1.05 mm. Using a cubic lattice as a basis, we further examine dimensional fidelity with respect to 2D lattice void dimensions and part position, finding differences between length scales and within parts, due to physical processing artifacts. Finally, we demonstrate a functional grading strategy based on process control methods to compensate for dimensional deviations. Using an iterative approach based on a naïve process model, deviation of the planar strut radius in a cubic lattice was decreased by approximately 85% after two iterations. These insights and strategies can be readily applied to other structures, characterization techniques, and additive manufacturing processes, thereby improving the exchange of information between digital and physical processing and lowering the energy barriers to producing high-quality lattice parts.

1. Introduction

3D printing and additive fabrication techniques have distinct advantages over conventional methods like injection molding, machining, and other subtractive techniques, which have opened up the possibility for new devices, applications, and effects brought about by so-called ‘additive-only’ structures. Lattices are one such feature, which may also be referred to as periodic cellular structures (open or closed), metamaterials, or architected materials. These structures are unique in their regularity and ability to create a macroscopic structure with properties that a solid part may not have [1]. Early lattice applications focused on making lightweight parts with the optimal distribution of material for withstanding forces specific to an application. As additive manufacturing has matured and computing power has become more accessible, new applications for lattices have emerged across a range of scientific disciplines. These materials have potential to make new devices tailored for specific applications including heat exchangers [2,3], catalyst supports [4], cell culture and tissue scaffolds [5–8], static and

flexible supports [9–13], consumer products, or otherwise. However, wide-spread adoption of lattice-based technologies remains challenged by a stepwise discretization in 3D printing processes that separates the digital design process from physical production processes. Fig. 1 illustrates the additive manufacturing process from intent to application. The application and additive manufacturing design constraints inform the exchange between the conceptual design space and software tools. Once the part goes to the printer, each physical phase can inform the digital phases in most applications only after completion, which makes this process necessarily iterative and time-consuming. Therefore, increased access to digital tools, materials processing knowledge, and the ability to translate physical results into digital inputs are necessary for the evolution of current and novel applications of complex additive-only architectures.

Challenges to the realization of functional lattices originate at the digital processing stage. Lattice generation software in general may comprise a variety of steps including primitive discretization, lattice population, trimming, functional grading, topology optimization, and

^{*} Corresponding author.

E-mail address: cfromen@udel.edu (C.A. Fromen).

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Boolean operations. Any trimmed struts are left as unsupported members of the lattice, having only one point of connection to the bulk structure. As a result, the lattice structure accumulates points of mechanical susceptibility, with hanging struts requiring support structures in most current additive manufacturing processes that contribute unnecessary material use and processing time to the economics of part manufacture. From a computational perspective, one solution is to use an adaptive meshing algorithm for generating conformal and optimized tetrahedral and hexahedral meshes, along with lattice structures [14–17]. This approach can be well-integrated with structural goals, but the demand for computational resources can be a burden, and variability in cell size and orientation can make these lattices unsuitable for applications which may require a regular internal structure at all but the boundaries. From a design perspective, one facile route is to use a solid skin. However, this leaves fewer paths for material flow during the print and post-processing steps, and it precludes processes which require ultraviolet (UV) curing steps after printing. An alternative to the solid skin is to remove hanging beams or connect them to other parts of the lattice. Yet this approach may limit the conformation to the original primitive or create regions of higher density while still leaving some regions that require additional support. While solvent-removable support material is a processing solution that mitigates the active post-processing burden, the need for multi-material printing may limit production scale. In effect, lattice structures that would enable novel applications and address the issues of material waste and post-processing are those which are fully conformal, uniform, and open. Lattice software from firms such as Carbon and nTopology can produce adaptive conformal lattices with advanced features. However, these types of lattices are not currently found in non-specialized software offerings.

Transferring the lattice file to the printer and discretizing the part into machine commands marks a significant transition from digital processing to physical processing and a wealth of additional challenges. The resulting physical lattice part is influenced by all of the conditions, controlled and otherwise, that arise from printing and post-processing steps. These include thermal gradients, material defects, chemical reactions, support printing and removal, solvent exposure, and model cleaning. All of the physical processing components contribute to the potential for missing beams, variable feature sizes, and surface characteristics which deviate from the intended CAD design [18]. These affect how well a part works in the application and the extent to which optimization can be effective in reaching design goals. Recent efforts have examined process artifacts and incorporated as-printed features like surface roughness and defects into finite element simulations to evaluate their impact on structural performance [18,19]. Further analyses have examined the deviation of printed parts at the intersection of beam lattices, which may accumulate additional material and contribute to differences in simulated and real behavior [20]. These materials are often characterized by X-ray computed tomography (CT), which offers a high-resolution 3D reconstruction of the part. This "digital twin" can be valuable for both process quality evaluation and more realistic simulation inputs [21]. Previous work has also used dimensional characterization to implement design-focused compensation methods to improve agreement between as-designed and as-printed parts [22]. Such characterization methods can be coupled with numerical models to facilitate model-based control of additive processes [23]. Despite these advances, new applications often rely on trial-and-error methods to identify

successful lattice design parameters and print settings, taken as any one of the semi-closed loops of the continuous cycle in Fig. 1 [24]. Unfortunately, the effective description of additive manufacturing processes, which is necessary to inform digital models and compensate for physical deviations *before* the parts reach the application, is a continuing challenge.

The barriers to broad adoption of lattices and additive-only periodic structures in general are many: researchers must be able to access lattice generation software with low risk, design with guidance on the differences between the theoretical and accessible spaces, and account for process conditions that impact print geometry. In order to address these barriers to entry and accelerate the transition of additively manufactured periodic structures from research to practice, we propose a pipeline for designing these lattices, assessing the accessible design space, and compensating for technological barriers that exists in generation and processing. In this work, we examine a facile, accessible process for taking periodic structures from concept to creation and addressing challenges associated with their development. We describe a framework for generating uniform lattices and conformal lattice skins without need of advanced meshing techniques. Printing these structures using the digital light processing (DLP) variant offered by the Carbon M1, continuous liquid interface production (CLIP™) [25], we observe that porosity and strut dimension are approximate predictors of print success across a range of lattice configurations that approach the printer resolution. Through design-stage functional grading, we address curing artifacts with a generalizable procedure that is broadly applicable for a range of parts and systems. In combination, these strategies reduce print waste, improve the dimensional fidelity of additively manufactured parts, and inform the digital lattice design process with physical results, strengthening the potential for fully open, additive periodic structures to expand into untapped applications.

2. Theory

2.1. Proposed lattice generation approach

The design process of periodic lattice structures for additive manufacturing has been discussed previously for various techniques [15,26,27]. Here, we adopt a number of the same conventions and expand upon them to demonstrate a facile and flexible approach to generating lattice structures that are conformal, uniform, and open. In our approach, lattice generation starts with a primitive shape that can be a boundary representation or mesh object, as is common for other procedures. The primitive is then discretized into structured hexahedral elements, referred to as voxels, which designate the spaces for unit cell tessellation. Within the space, voxel dimensions can be uniform (cubic) or non-uniform in any of the 3 directions (prismatic). Furthermore, the origin of voxelization has implications for lattice generation. One method, which grants 2D symmetry, is to set the origin at the top or bottom extreme of the primitive and at the center (or centroid) of the 2D projection onto that plane. The other, which grants 3D symmetry for a symmetrical primitive, is to set the origin at the center or centroid of the primitive volume. Our process employs the latter and uses a parallel approach for internal lattice and skin generation. Generation of the internal structure consists of full voxelization, unit cell population, and inclusion. Generation of the conformal, open skin consists of boundary

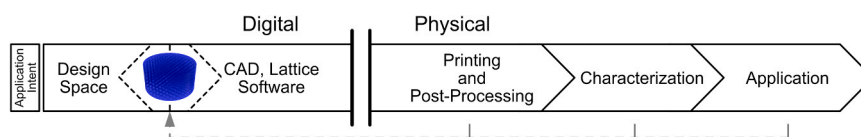


Fig. 1. Simplified lifecycle of lattice parts from concept to application. Digital processing is initiated by the application intent and informed by process constraints. Lattice generation can occur fluidly through CAD modifications. After physical processing begins, each step in the cascade will typically reach completion before information can be passed to another phase for a new iteration.

voxelization, unit cell connectivity description, and intersection. Finally, the internal structure and skin are combined for generation of the final mesh, and in symmetrical cases, the final part can be used in an orientation-independent manner. This entire process addressing the digital lattice software design is depicted in Fig. 2. While the concepts described here are intended for an open lattice, the procedure is appropriate for combination strut-surface lattices, solid skins, and solid-lattice hybrid structures.

Population of the internal structure is shown in Fig. 2A. To fill the primitive with the lattice, the unit cell is transformed from its bounding box to the new set of bounding boxes defined by each voxel from the primitive discretization. All of the internal structures are mapped to the new space at each discretized point (Fig. 2B). For a coarse first pass, the voxels contained within the primitive and intersecting its boundary are kept for the lattice. In the inclusion step, shown in Fig. 2E, the struts crossing the primitive bounds are truncated at the boundary. It is important to note that based on this scheme, the voxelization strategy does not adapt to the local contours of the primitive shape. As a result, boundary voxels must be truncated by the part geometry. Many parts used in practice may require a fully conformal voxelization approach based on adaptive elements, but this is not a necessary condition for generating a conformal lattice. The distinction is that in this scheme, the internal structure is uniform, and the peripheral elements are of a reduced volume. This allows the lattice boundary to still comply with more complex structures, regardless of whether they take on a shape that is simplified, regular, or amorphous. For cases requiring homogeneity in element dimension or directionality, an adaptive scheme may be worth the computational resources. However, in cases where the truncated region is small in comparison to the inner structure, these edge effects may constitute a negligible deviation from the bulk lattice properties.

The open skin generation process consists of the three steps shown in Fig. 2B, D, and F. First, a set of surfaces must be defined for the unit cell connectivity. These surfaces describe the relation between any struts that should be connected in the open lattice skin. A simple way of constructing these surfaces is to create a solid bounded by the struts as

edges. However, for consideration of the print process and orientation, additional surfaces should be included such that possible overhangs and discontinuous segments are supported. These mapped surfaces are then intersected with the primitive to create a lattice skin that perfectly aligns with the trimmed beams from the full voxel population. Finally, a set of beams is created from the union of the internal lattice, lattice skin, and relevant wireframe elements from the CAD part, which includes edges of the original geometry that should be maintained in the lattice. Once the lattice skeleton has been generated, the struts can be meshed uniformly or with functional grading, and the part is prepared for Boolean operations. These can include combination with selected solid regions or trimming a base parallel to the build platform, enabling support-free fabrication.

This approach is advantageous because it does not require *a priori* description of the overall lattice topology or mesh connectivity and it maintains the overall uniformity of the lattice structure. It does not depend on the quality of the input structure, as may be the case for adaptive meshing techniques, and multiple structures can be quickly generated from a single tessellation of the original lattices. It is also not necessary to adaptively find nearby nodes or vertices for connectivity, which may be subject to various tolerances and produce different behavior, depending on unit cell topology. Furthermore, it eliminates the need to support the free beams manually or through software scripting, and since the lattice is self-supporting, there is no need to support the lattice internally or at the periphery, which reduces material usage and post-processing steps. The simplicity of the method also enables symmetry-based lattice generation for parts with n-fold symmetry, where a subset of the entire lattice can be generated, trimmed, and propagated to create the final print file.

2.2. Process control

Increasing attention is being given to real-time process monitoring and closed-loop process control during part manufacture to enhance part quality and repeatability in all facets of additive manufacturing [28]. A lack of fundamental understanding of many of the physical

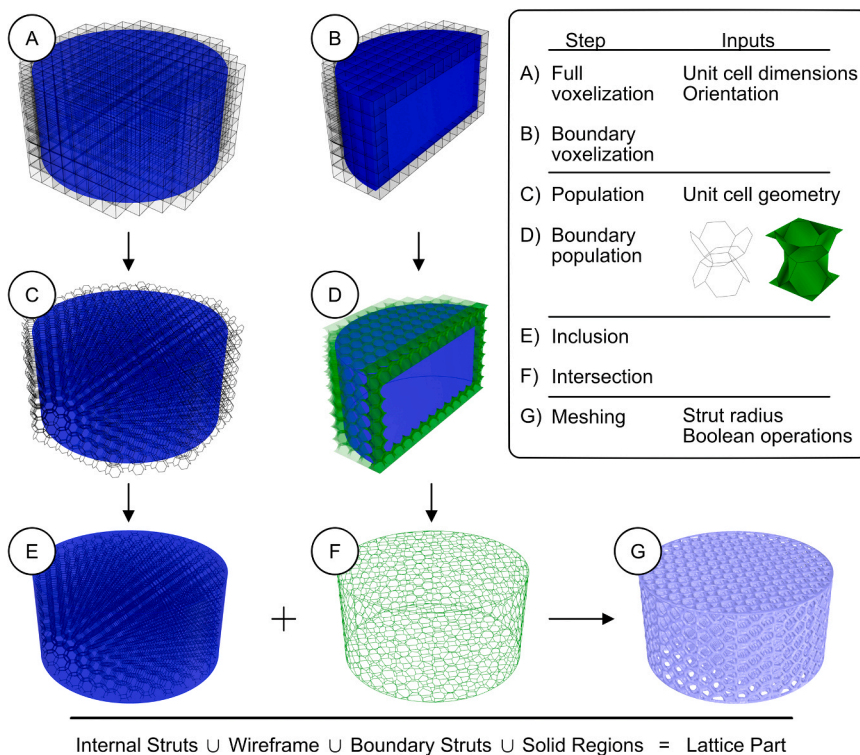


Fig. 2. Representative schematic of lattice generation procedure described in Section 2.1. The design space primitive is subjected to full voxelization (A) and boundary voxelization (B). A defined unit cell geometry, along with a description of the node and strut connectivity, is used to populate the full (C) and boundary voxels (D), filling the primitive model with lattice beams. Beams in the full lattice are trimmed (E), and the boundary connectivity is intersected with the primitive (F) to produce the main lattice body. Finally, the combined trimmed lattice and skin is inflated and combined with other part geometry to generate a triangular mesh for 3D printing (G). The lattice part consists of a combination of internal struts, boundary struts, CAD input edges (wireframe), and solid regions.

processing steps has led to on-going efforts to develop predictive process models for a variety of additive processes [23,29]. Efforts to generate real-time, closed-loop control over the various processes focus on the development of *in situ* measures of the physical part generation [30,31]. Lacking physical models and real-time monitoring controls for existing processes, part successes, especially in the early prototype phase, rely on heuristics and trial-and-error approaches. This is especially detrimental to printing additive-only structures such as lattices, especially as critical feature sizes needed for the end application approach the resolution of the feasible design space. Real-time corrections addressing deviations occurring between the digital design process and the physical production processes for lattices remain an open challenge.

The case presented here by the iterative digital and physical lifecycle (Fig. 1) is a batch process without a well-defined process model or integrated metrology for real-time measurement of disturbances and printed dimensions. This limits the potential for direct use of feedforward or feedback control [32]. In the case of batch chemical processes, a scheme like iterative learning control (ILC) can be well-suited [33]. However, improved methods still depend on process models and mathematical filters, which are likely undetermined for a novel system such as a first-time print or prototype, and the number of iterations required for sufficient set-point tracking may exceed time for development in early stages. Furthermore, because curing phenomena are expected to be a complicated function of interactions between process conditions and input dimensions, the learning filter operation may require tuning for every unique configuration. In light of this, the combination of ILC's iterative form and the compensation logic of feedforward process control is an interesting scheme for study. For demonstration of the lattice generation process and characterization techniques, the approach in this work assumes each iteration to be defined as a unique process. The controller for the following iteration is defined by a simple set point tracking controller, the reciprocal of the process function input-output map, which should eliminate the error after a single iteration for an ideal case [34]. The results of this approach are examined in Section 4.4.

3. Materials and methods

3.1. Computational and software resources

CAD models were generated using Rhinoceros 3D 6 and Grasshopper (Robert McNeel & Associates) on a desktop computer running an Intel® Xeon® E-2126G processor with 64 GB RAM. This software was selected for its accessibility and extensibility because it supports visual and text-based scripting, along with a large library of community-supported packages. An additional benefit is Rhino's relatively low cost of entry (a one-time purchase) and the free, open source availability of many of these plugins. As a result, any user can create latticed parts in the same software used to define the native CAD geometry. In this work, lattices were created with internal Grasshopper and GhPython scripts by adapting select components of the community lattice plugin, Crystallon ($f=f$). The combined lattice curves were converted to meshes using the Dendro plugin, which provides a marching cubes implementation (ECR Labs). Further geometry processing, analysis, and data processing were performed using internal scripts before exporting as stereolithography (.stl) or 3D Manufacturing Format (.3mf) files and sliced using the Carbon cloud platform (Carbon Inc.). Data were imported from Grasshopper and FLJI [35] into R v3.6.3 (R Foundation for Statistical Computing) for further processing and representation. A list of R packages used is included in Supplemental Section 1.

3.2. Lattice generation

The lattice design approach described in Section 2.1 was used to generate a minimal lattice example from a rounded rectangular prism with arbitrary but unequal edge lengths. The parameters used for

generating the lattice were selected to produce a trimmed $3 \times 3 \times 3$ unit cell array. Standard beam lattice geometries included Cubic, Kelvin (truncated octahedron), body-centered cubic with XY connections (BCCxy or vertex octahedron), and Rhombic Dodecahedron (RD) from the Crystallon plugin, and the Weaire-Phelan (WP) unit cell topology was modeled based on the volume-filling structure provided by the BullAnt plugin (GeometryGym). Two additional arbitrary cells, dubbed sheared cube and taut cube were generated internally to illustrate the flexibility of the lattice generation methodology for curvilinear shapes.

3.3. 3D printing

All parts were printed using a Carbon M1 3D printer with 100 μm slicing in the vertical print direction. Both one- and two-part resins were used in this work to evaluate as-printed and as-cured mechanical properties: urethane methacrylate (UMA 90) in cyan and black, elastomeric polyurethane (EPU 40, black), and silicone urethane (SIL 30, gray). UMA 90 is a one-part resin which prints and cures stiff by exposure to ultraviolet (UV) light only; EPU 40 is a two-part resin which prints soft and retains its flexibility with slight stiffening after thermal curing; and SIL 30 is a two-part resin which prints semi-stiff and becomes flexible after thermal curing. All parts were processed within the manufacturer's recommendations, which entailed solvent exposure with isopropyl alcohol (IPA, 2-propanol, Certified ACS or Optima™ for HPLC, Fisher Chemical) to remove residual resin after printing (less than 15 min for UMA, less than 1 min for EPU, and less than 5 min for SIL 30) and drying with a stream of compressed air before UV curing at 5–10 mW, 365 nm (Opticure LED Cube, APM Technica AG) or thermal curing (DKN602, Yamato Scientific America Inc.).

3.4. Lattice models

Lattice models were designed based on a primitive input, cubic unit cell length (l_c , mm), and strut radius (r , mm). As a precursor to strut radius, the dimensionless strut radius ($r^* = r / l_c$, unitless) was defined to facilitate comparison between unit cell length scales (Section 4.2). This quantity was then converted to the absolute radius value for input to the mesh generation step and used for comparison between lattice configurations. Where reported, porosity was defined as the ratio between the volume of a repeat lattice unit within the unit cell bounds and the volume of the unit cell. These values were calculated based on the CAD geometry and expanded upon previously determined values [36]. To identify the boundary between printable and non-printable lattice configurations, a range of lattice structures was designed around the printer's recommended feature sizes. A small lattice array consisting of all combinations of cell length (1.5, 2, 2.5, 3 mm) and dimensionless radius (0.05, 0.07, 0.09, 0.11, 0.13) was generated for Cubic, Kelvin, BCCxy, and RD unit cells. The WP array was generated with an additional cell length at 3.5 mm. Each lattice generated for a cylindrical primitive (5 mm radius $\times$ 15 mm height) patterned in a 14 mm $\times$ 14 mm cell grid, with a support base that permitted printing all parts together as a single piece to facilitate accurate identification each lattice configuration and post-processing. All parts were printed in UMA 90 Black under standard print settings (100 μm slicing, other conditions set by manufacturer software) and washed for approximately 5 min in IPA while agitated at approximately 80 RPM.

A larger lattice array was generated for all combinations of cell length (0.5, 0.75, 1, 1.5, 2, 2.5, 3, 3.5 mm) and dimensionless radius (0.11, 0.12, 0.13, 0.14, 0.15) to test print success in different photopolymer resins. This array was generated for only the cubic unit cell geometry, with the primitive and spacing in the same configuration as the small arrays. This array was used for printing in UMA 90 Black, SIL 30, and EPU 40. Two additional arrays were generated for the following cases: for EPU, an array without a flat base was generated after the forces associated with printing caused the full print array to delaminate from the print platform; and for UMA 90 Black, a randomized array was

generated where the position of each lattice configuration was assigned randomly before printing, as a control. For single-part resin arrays, print success was determined by visual inspection before curing; a part was classified as failed if it delaminated from the platform or lost structural integrity of the core shape (e.g., frayed beams, collapse, or damage after contact with another part). Printability for two-part resins was determined by visual inspection after thermal curing.

Dimensions and images of the lattice arrays are included in [Supplemental Section 2](#).

To investigate dimensional fidelity in representative parts, six cubic unit cell lattices were designed with arbitrary unit cell lengths in the printable range, with a constant dimensionless radius. The parts consisted of 33 mm diameter $\times$ 10 mm height cylinders designed with unit cell sizes in the range of approximately 1–4 mm at a constant r^* of approximately 0.14, corresponding to 5 cells-per-inch lattice, as demonstrated by Klumpp et al. to represent a real system used in alternative applications for additive manufacturing [37]. The window sizes of these lattices included: 0.74 mm, 0.95 mm, 1.3 mm, 1.86 mm, and two additional parts used to impose a symmetrical print layout. Additional lattice dimensions are listed in [Supplemental Section 2](#). The lattices were loaded on the platform in equal spacing, to fill the available print area for a single run. Parts were printed in UMA 90 Cyan, a variant of UMA 90 with a different color additive. Post-processing, parts were washed in IPA for approximately five minutes while agitated at approximately 80 RPM before curing according to manufacturer recommendations. After curing, parts were allowed to equilibrate at least one day before imaging.

3.5. Dimensional characterization

All imaged parts were loaded onto a polypropylene tray and imaged on a BioTek Cytation 5 Multimode Imager (BioTek Instruments Inc.) under 4X magnification. The sample images were constructed from a series of bright field images stitched using the instrument's default settings. The resulting images were imported into FIJI for image analysis, where the regions of interest were selected by automatic thresholding in FIJI [35]. Window area (A_w , mm²) was measured using FIJI's built-in tools and normalized by the designed window area, calculated as the square of window length (A_{design} , mm²). Of the lattices analyzed in the scaled window size array, the 0.95 mm window was selected for investigation of the effects of platform loading and position. This choice was made based on the size approaching the scale with the greatest degree of overcure but less significant curing at the core; it also contained a sufficient number of windows to characterize the behavior. Lattices were printed in UMA 90 Black in two print configurations: (1) with a single part at the center of the platform, and (2) with three parts evenly spaced on the platform, spaced at a center separation distance of 50 mm. Both configurations were printed in $n = 3$ replicates for statistical evaluation. Parts were again washed in IPA for approximately five minutes while agitated at 80 RPM before curing and leaving to equilibrate at least overnight before final window quantification. Results were compared statistically using analysis of covariance (ANCOVA) by fitting a linear model of normalized window area (A_w/A_{design} , unitless) with radial distance from the center of the lattice (R , mm) as a covariate. A subset of the data ($R < 14$ mm) was analyzed to avoid possible edge effects from the imaging setup and to remove truncated edge windows from the analysis. Window sizes were selected within a reasonable threshold to remove aberrant data points generated from image stitching or FIJI analysis.

3.6. Dimensional compensation

Using the window dimensions of the single centered part from the platform loading trials, a first-order linear model was generated for the approximate strut radius based on the ideal relationship between window area and strut radius ($r_{\text{approx}} = (l_c \sqrt{A_w})/2$). From the linear model,

the slope was used to estimate the effect of processing on strut radius as a function of radial position in designing a compensation method for curing effects. A single compensator was generated as the inverse of this relationship and used to specify the strut radius in a functionally graded lattice, which was printed and processed under the same conditions as the original ($n = 3$ replicates). The window area results after the first compensation stage were used to design a second compensator for another iteration ($n = 3$ replicates). All parts were printed using UMA 90 Black, following the previous printing conventions.

4. Results and discussion

4.1. Lattice generation

The lattice generation process for a Kelvin unit cell and a cylindrical primitive are shown in [Fig. 2](#). Following the digital processing steps outlined in [Fig. 2A–G](#), a fully conformal, uniform, and open cylinder was generated, with full connectivity and no overhangs at the boundary. To illustrate the robustness and broad applicability of the lattice generation technique, [Fig. 3](#) shows trimmed lattice results for a $3 \times 3 \times 3$ unit cell array. For all unit cells, the generation process produces a fully connected network between the internal lattice structure and the open skin at all of the primitive faces. Each lattice maintains its connectivity at the rounded corners, which shows the potential of this approach to generate conformal skins for uniform lattices of any cell length. Furthermore, each of the unit cell edge lengths can be defined independently and without relation to a parameterized design space, while still retaining the self-supporting structure for an arbitrary curved primitive.

Standard beam lattice geometries are normally considered as node-strut combinations. For such standard beam lattice structures, including cubic ([Fig. 3A](#)), BCCxy ([Fig. 3B](#)), RD ([Fig. 3C](#)), and WP ([Fig. 3D](#)), the basic connectivity can be considered as the bounding envelope of a solid prism enclosed by the strut beams, trimmed at the unit cell faces. The prism-based approach is implicit in many stochastic lattices that mimic trabecular bone structures [6,15]. Described in more detail and for a generalized case, as in [Section 2.1](#), our lattice generation approach can expand upon conventional node-strut architectures. This approach is well-suited to truncated shapes and more complex structures that would be observed in triply periodic minimal surface structures or curved beams, as exemplified by the sheared cubic ([Fig. 3E](#)) and taut cubic ([Fig. 3F](#)) unit cell geometries. In both cases, the generated lattice is populated uniformly within the primitive and connected by the net skin at all trimmed points. However, with curved-beam unit cell geometries, care must be taken to design the curvature with physical processing in mind, namely the print direction and support orientation. In order to create a self-supporting structure, the taut cube was designed with additional supports in the unit cell wireframe to avoid creating unsupported regions from up-turned beams along the print axis. Its connectivity description required interior surfaces to ensure that these regions remain connected in the lattice skin, which can be seen in the final output. Depending on print conditions and the degree of curvature relative to the unit cell features, connectivity descriptions for standard geometries may also require additional features to support all regions.

As a result of this generalized approach, the open skin address multiple challenges in lattice design for additive manufacturing, which have been described in the previous work on net skin generation by Aremu et al. [27]. Self-supporting lattices described previously use surfaces or closed cells for structural support [38,39]. While curved regions in triply periodic minimal surfaces and other surface lattices are still self-supporting when trimmed, beam lattices may leave hanging struts that must be supported manually or programmatically by identifying print “islands” and connecting them to the print base or other nearby struts. Some software solutions can remove or connect hanging beams, but this limits the ability of the lattice geometry to conform to the parts used in application, potentially sacrificing structural integrity or lattice features critical for the application. Generating a solid skin

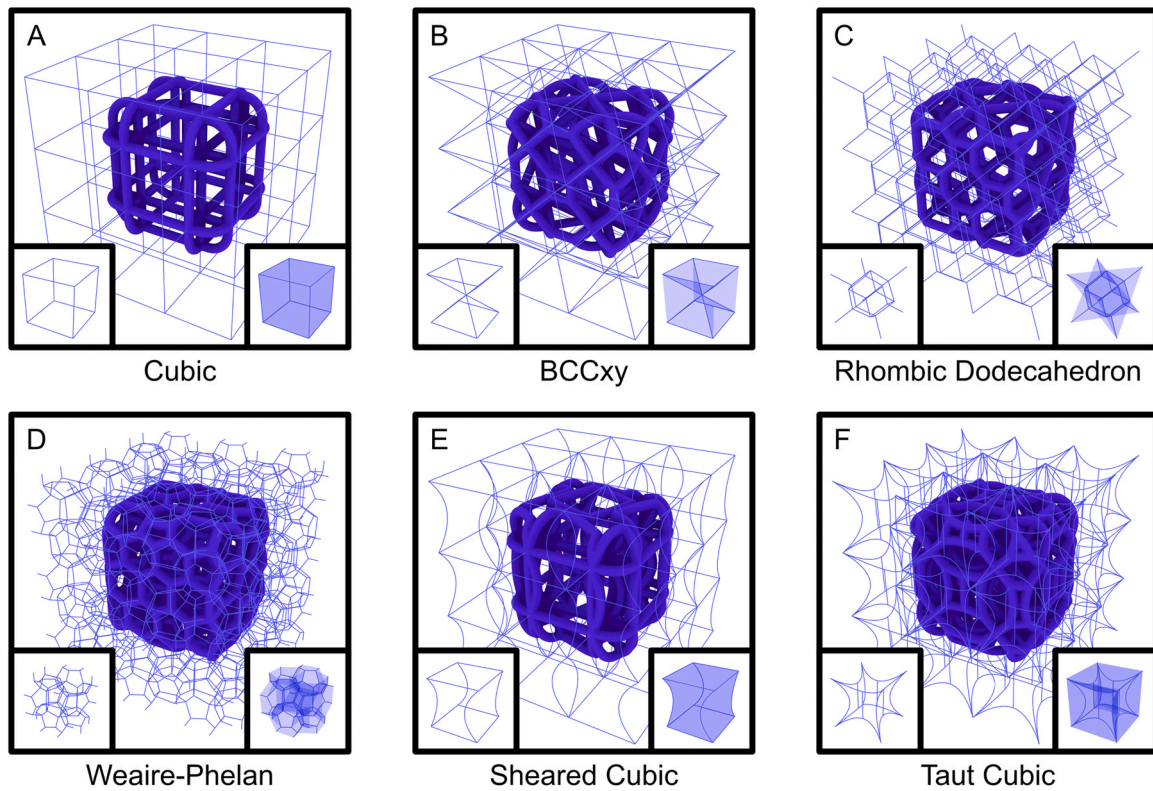


Fig. 3. Visualization of unit cells and trimmed lattice structures for the geometries investigated and two demonstration geometries: (A) Cubic, (B) BCCxy (C) Rhombic dodecahedron (RD), (D) Weaire-Phelan (WP), (E) Sheared cubic, and (F) Taut cubic. Each panel includes the following: unit cell skeleton (inset, left), unit cell connectivity (inset, right), and tessellated unit cell and uniform lattice with skin (main). The rounded corners of the rectangular prism highlight the flexibility of the system to generate a uniform lattice with an open skin that conforms to an arbitrary boundary.

may be a viable alternative for some processing methods and it results in increased structural integrity at the lattice boundaries and mating surfaces. Despite the structural benefits, a solid skin can make effective support removal difficult to impossible and it may increase post-processing time for powder and resin 3D printing processes. This not only increases overall production time but also makes certain configurations unattainable if the processing steps or solvent exposure time surpass permissible limits for the specified material properties.

There are some subtle differences between this approach and the orthographic voxel projection method proposed by Aremu et al., which served as inspiration for this work [27]. First, our presented method does not require voxelization at the level of the lattice beam to generate the skin, which can facilitate a skeletal description of the lattice network before applying a thickness; this may have implications for manufacturing techniques which translate geometry to tool paths, rather than rasterized planes. Second, open regions within the lattice network are maintained at the boundary since the process here does not look for proximal points within the lattice to project onto the skin region. Third, this method is orientation-agnostic; aside from the initial discretization, it is not contingent upon a given coordinate system or a defined axis along which elements should be projected. This can create a challenge with curved geometries like in Fig. 3F. Such geometries require careful description of the support structures in the unit cell and connectivity. The example here should be modified *a priori* or during generation if the print orientation would create unsupported regions at local extrema. Regardless, this generation approach can be applied to arbitrarily defined beam lattices, mathematically defined structures, and even combination strut-surface lattices by merely describing the connectivity of the unit cell geometry. The advantages of an open structure go beyond reduced waste and processing time. They enable a broader range of processable structures, facilitate lattice interchangeability in production assemblies, and support high-throughput lattice screening.

The efficiency of this approach further reduces technological barriers to adoption and investigation, expanding the potential for utility of this material class and the horizon of untapped applications.

4.2. Printability

Given the range of printer specifications and material properties available for 3D printing, it is important to benchmark the print process and determine the dimensional space available for printing. The lattice parts described in Sections 2.1 and 4.1 print successfully, as shown in Fig. 4A and B respectively. An additional example showing a lattice for more complex part geometry is shown in Fig. 4C. The part shown is a model of the Taut Cubic geometry from Fig. 3F in the conformation of the commonly used GE engine bracket [40], with the voxelization plane (0.66, 0.5, 0.56) offset from the print plane (0, 0, 1). This highlights the flexibility of this approach and demonstrates the capability to use it with curved beams, complex geometries, and hybrid solid-lattice structures.

To identify broader trends in lattice printability, the five standard unit cells described in Section 3.2 were printed in a small array of cell lengths and strut sizes. The printability results of the lattice arrays are shown in Fig. 5. Print success is indicated for each combination of dimensionless radius (r^*) and cell length (l_c), with red shaded regions indicating part failure. For images of the printed parts, see Supplemental Section 2. As indicated by the stairstep pattern, successful lattice prints depend on larger relative strut diameters at smaller unit cell length scales. Cubic, BCCxy, and RD lattices showed similar failure trends, while Kelvin and WP geometries showed fewer failures by comparison. Where the recommended positive feature sizes in the XY plane and Z direction are indicated by the dashed and solid lines, respectively, it is clear that parts at or below the minimum Z feature size fail, while parts with dimensions between the manufacturer's recommended sizes may or may not print successfully, depending on the relative strut radius.

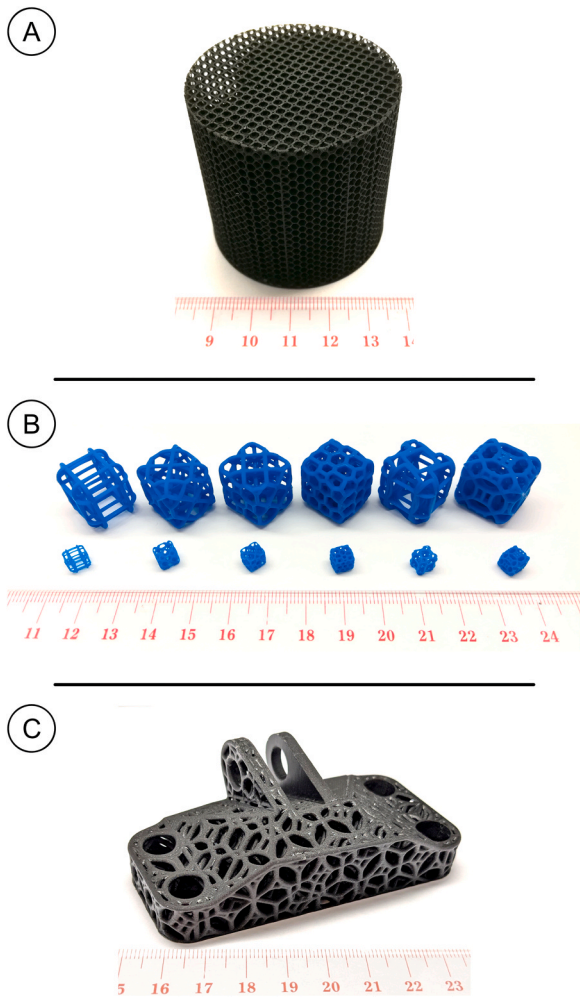


Fig. 4. Photographs of printed parts, adjusted for visual clarity. (A) Larger scale lattice of the design in Fig. 2G, used in fluid applications [36]. Approximate dimensions: 50 mm diameter x 50 mm height; $l_c = 2.4$ mm; $r^* = 0.11$. (B) Lattice designs from Fig. 3A–F, from left to right. Approximate dimensions: 5.5 mm x 6 mm x 6.4 mm (small); $l_c = 3$ mm; $r^* = 0.07$; large parts are scaled linearly by 3x. (C) GE engine bracket consisting of internal struts, boundary struts, wireframe edges (left, front), and solid regions (right, back). Unlike in solid skin lattices, print material, solvent, and air can pass freely throughout these open lattice parts.

Parts with dimensions above both recommendations print successfully throughout.

Because all parts were designed with the same strut and unit cell dimensions, they each are similar in length, but they may have different porosities or relative densities depending on the total strut volume within the boundary of the unit cell. As a result, some lattices use more material and may promote more undesirable phenomena in the curing

process, leading to beam fusion and unintentional thickening or closed voids where beams are closer than a process-influenced threshold. Minimum and maximum features sizes for lattice structures have been described numerically for powder fusion processes [41], where limitations are imposed by the size of the powder substrate with respect to lattice openings, but it is clear from the phenomena observed in this case that physical processing effects still impact processes which use liquid material that theoretically should be able to permeate any open void space in the print process. Another common property between the lattices examined here is that they are all bending-dominated structures by Maxwell's stability criterion. The distinction between bending- and stretch-dominated structures is a key selection parameter when choosing a lattice for a given application. For example, if the lattice should provide rigidity, a stretch-dominated structure may be better-suited; but if the lattice should provide cushioning, a bending-dominated structure would deflect more easily. This characteristic is not a focal point of this work, but it should be noted that bending-dominated structures exhibit lower structural efficiency compared to other geometries with more interconnected beams [42]. This can have implications for the printability of the lattice geometry itself, if the beams of the lattice are poorly suited to sustaining a load without deflecting. However, it has been previously demonstrated that an open skin can still provide structural support, especially with increasing thickness of the skin [43]. In this study, a uniform strut dimension was selected to provide results for a representative case of minimal modifications, so this effect was not examined in detail.

With these considerations, the cubic unit cell was selected for further analysis in the remainder of this section and Sections 4.3–4.4. This decision was made to facilitate print design, evaluation, and visual characterization. The cubic lattice accommodates a wide range of dimensional parameters for a relatively low computational burden and many of its features exist in the same plane, expediting image-based dimensional validation. The printability results for cubic lattices printed in a larger range of length scales for cell length and strut radius, in addition to different resins, are shown in Fig. 6. For UMA 90 Black at the bottom of the printed range, the stairstep trend is maintained up to a critical value at approximately $r = 0.065$ mm. In the UMA 90 array, two parts were categorized as failures after printing. The lattice at (0.5, 0.13) failed after being dislocated by another failed print, and the lattice at (1, 0.11) was classified after a loss in structural integrity at the periphery. Considering the influence of position and proximity to other lattice parts, the UMA 90 array was printed again in a shuffled order, and it followed similar trends as the ordered array. Notably, (0.5, 0.13) and (1, 0.11) printed successfully. This indicates that part proximity may have an impact on lattice printability during printing and post-processing, which is an important consideration for both lattice design and processing simulation. Unlike the smaller lattice array, printable lattices do exist below the minimum recommended Z dimension, but again this requires an increase in the relative strut radius at smaller cell length scales.

For SIL 30, nearly all parts are below the minimum recommended XY and Z dimensions, but over 75% of the tested parts print successfully.

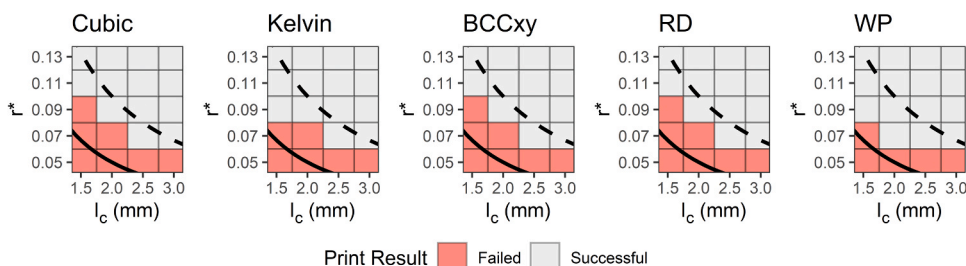


Fig. 5. Representation of the printability results for the 20-lattice arrays printed for each of the five unit cell geometries in UMA 90 Black. Highlighted regions indicate part failure. The solid line indicates the manufacturer-recommended minimum Z feature size, and the dashed line indicates the manufacturer-recommended minimum XY feature size. Points along the curve indicate equivalence between the minimum feature size and the corresponding lattice strut diameter. Full dimensions for all lattices and additional results for WP included in Supplemental Section 2.

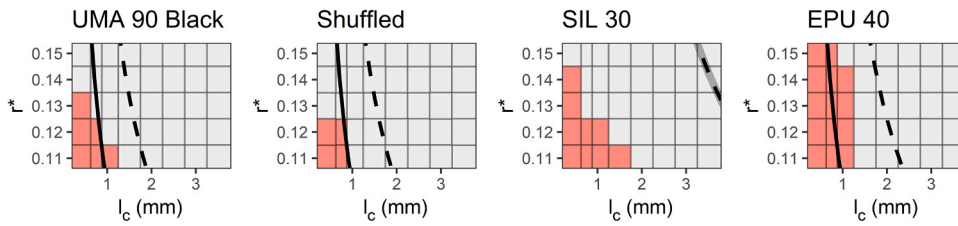


Fig. 6. Representation of the printability results for the expanded 40-lattice arrays printed for the Cubic unit cell geometry in UMA 90 Black, SIL 30, and EPU 40, including the randomized-order UMA 90 Black lattice array. Highlighted regions indicate part failure. The solid line indicates the manufacturer-recommended minimum Z feature size, and the dashed line indicates the manufacturer-recommended minimum XY feature size. Recommended feature sizes for SIL 30 are the same

in both orientations, as indicated by the semi-transparent solid line and dashed line. Points along the curve indicate equivalence between the minimum feature size and the corresponding lattice strut diameter. Full dimensions for all lattices included in [Supplemental Section 2](#).

Despite their successful printing, certain SIL 30 lattices may exhibit curing phenomena that causes the parts to fill the void space with cured resin, decreasing the as-printed lattice porosity and deviating from the design value. Resin-dependent differences seen here are consistent with observations by McGregor et al. [44] who showed that between two-part resins on a similar additive manufacturing platform, part dimension statistics varied between resin chemistries for otherwise identical designs. From our study, an additional insight is that the number of failures increases with decreasing stiffness of the as-printed part, where UMA 90 is the stiffest as-printed material, followed by SIL 30, and EPU 40. These results suggest that even with the benefits of support from the lattice net skin, the lattice design alone is not a sufficient criterion to predict success.

For comparison between length scales and unit cell geometries, the porosity for each lattice configuration was calculated based on the mesh generated for a single repeat element. The culminated results for prints in UMA 90 Black were used to generate a Voronoi diagram relating printability to cell porosity and strut radius (r , mm), shown in [Fig. 7](#). In this representation, the points correspond to tested configurations, while

the bounded regions are a key feature of the Voronoi diagram. Within the bounds is the space in closest proximity to the central point exclusively and the boundaries are equidistant to all nearest-neighbor points. This provides a facile approach to generating a pseudo-phase diagram of print success and failure within the experimentally sampled space. Print failure is indicated by red shaded regions. With some exceptions, lattice prints begin to fail below a strut radius of 0.15 mm and a porosity greater than 0.75 (relative density < 0.25). Regions far from printed points should be interpreted judiciously, given that the diagram is not a statistical representation or prediction of the results.

The theoretical minimum strut diameter should equal the largest dimension of the print resolution, which may be dominated by the XY resolution in the slicing plane or the slice thickness. In the present case, this theoretical minimum strut diameter corresponds to the slice thickness at 100 μ m, or a strut radius of 50 μ m (0.05 mm). The results of the combined printability arrays indicate that only certain unit cell geometries, like the cubic unit cell, are able to print near this length scale. In comparison with the other unit cell geometries presented, the cubic unit cell has the greatest porosity, and all of its struts are normal or

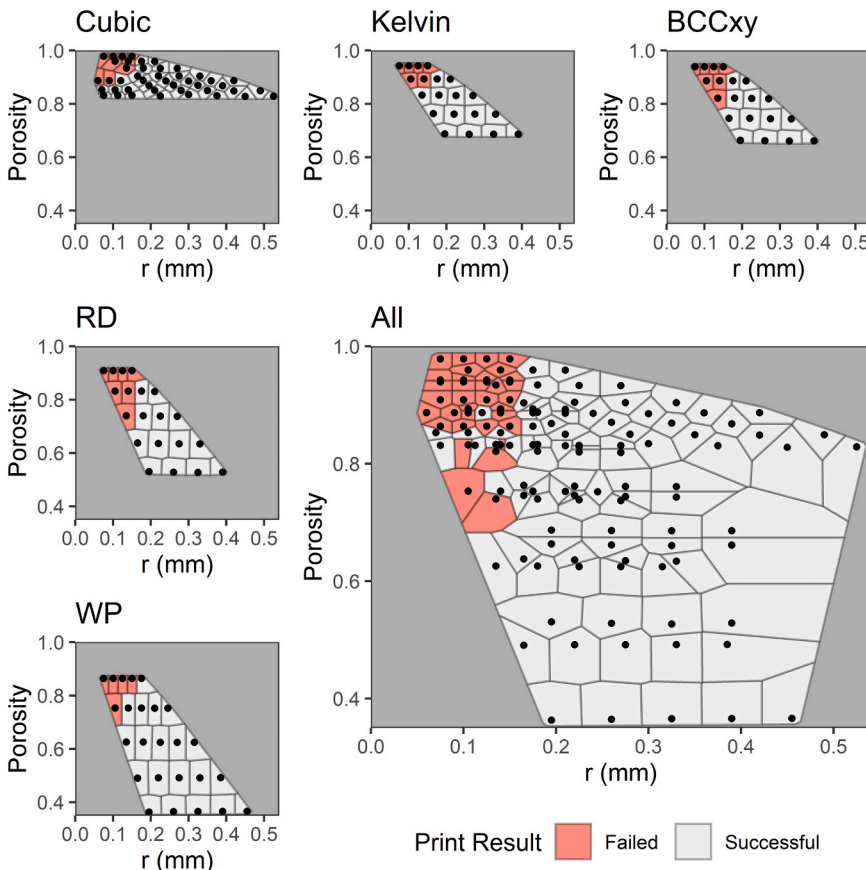


Fig. 7. Voronoi diagram of the printability results for all ordered UMA 90 Black lattice arrays. The Voronoi diagram range is restricted to the convex hull of all examined points. Points correspond to porosity and strut radius measurements based on CAD-generated lattices. Boundaries shown are mathematical constructs representing equidistant positions between neighboring points. Highlighted regions indicate part failure. Trends indicate part failure at higher porosity and lower strut radius, approaching the minimum resolution of the printer settings at 75 μ m x 75 μ m x 100 μ m.

orthogonal to the print direction. While there are horizontal struts, which goes against manufacturer recommendations, they are all bridged and supported by fully vertical struts.

Data generated in the printability arrays (Figs. 5–7) were obtained from a rapid screening process. The results suggest that lattice parts are scalable to an upper porosity limit and lower strut radius limit, which may differ from manufacturer recommendations. These results are confounded by differences in unit cell geometry, and resin type. Relative location of the part also exhibits influence over print results, and further investigation is necessary to confirm behavior in parts with larger overall dimensions (Section 4.3). While the printability array may not be representative of end-use lattice applications or all possible configurations, it is a valuable tool to benchmark the print system and is applicable for other printing technologies. Because not all systems may have pre-defined recommendations for feature size, the ability to quickly screen lattice designs facilitates the input of process-informed dimensional limits in latticing, topology optimization, and generative software options. This can focus resource use on designing for the space accessible to the process and limit the frequency of failed parts. Even when recommended sizes are available, it is possible that some parts are able to print at dimensions below the recommended features as a result of the unique structural properties of lattices. One limitation of this assay is the difficulty in identifying which parts may exhibit curing effects that would close pores. Where some lattices can indicate this by mere visual inspection, a non-destructive imaging technique would be required to determine whether the cells at the lattice core are open or closed [21]. Regardless, these results indicate that a wide range of lattices can be scaled linearly, even when their dimensions approach the bounds of hardware-limited or manufacturer-recommended dimensions. This further demonstrates the flexibility of additive manufacturing systems and the potential for a single device to fabricate parts with feature sizes up to an order of magnitude in difference, making them capable production tools for a range of applications.

4.3. Dimensional characterization

After observing a range of curing effects between different parts in the lattice printability array, two experiments were conducted to quantify these effects and test printability of parts at larger overall dimensions. First, an array of latticed parts was printed in UMA 90 Cyan resin in a single print and imaged for dimensional analysis. Each part had the same overall dimensions and lattice r^* , making the total part volume and maximum cross-sectional area comparable between all of the lattice length scales investigated (see Supplemental Section 3). Fig. 8 shows reconstructed micrographs and normalized window area as a function of radial position for three selected lattices. All of the parts examined had median normalized window areas below the design value, as shown in Table 1. By visual inspection of the as-printed parts, there was a noticeable radial trend for the smallest window size (0.74 mm). This was confirmed by image analysis. The spatial distribution of normalized window area showed that the smallest windows were concentrated at the center of the part for all configurations. As detailed in Table 1, the radial trend increases from less than 1%/mm to almost 4%/mm with decreasing l_c . The theoretical central window area ranged from approximately 85% to approximately 52% of the designed value.

The decrease in window area is indicative of an overcuring effect that is commonly observed in DLP 3D printing, occurring as a result of a variety of interacting phenomena related to the photopolymerization process [45–48]. Since a latticed geometry has many internal boundaries, the curing deviations are visible throughout the part and not only at the periphery. McGregor et al. previously observed radial dependence of as-printed part dimensions for two-part resins and attributed the phenomena to post-processing and thermal curing [44]. In their later exploration of hardware-related dimensional variability for the same resin chemistry used here (UMA 90), they did not observe a relationship between radial distance and feature dimensions [49]. This is in contrast

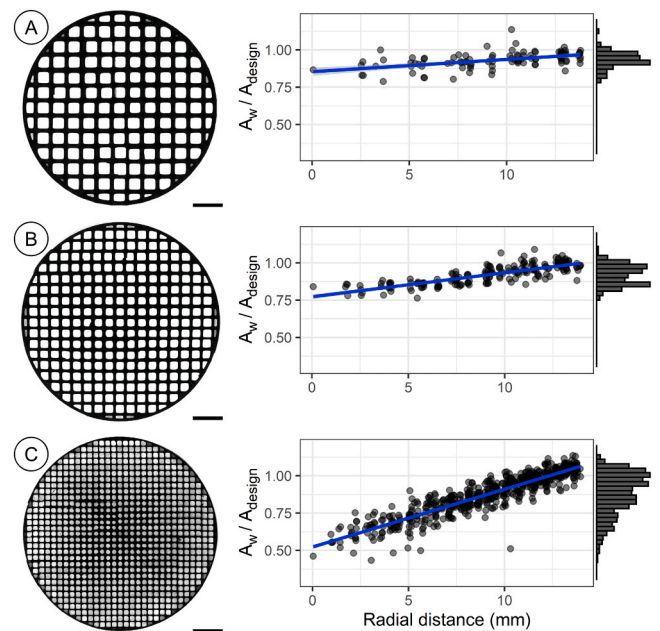


Fig. 8. Dimensional characterization of parts with different unit cell scale, fabricated from a fully-loaded print (3 selected parts, $n = 1$). From left to right: reconstructed micrograph, normalized window area as a function of radial distance from the part core, and histogram of normalized window area. (A) 1.86 mm window length, (B) 1.3 mm window length, (C) 0.74 mm window length. Scale bar is 5 mm.

Table 1

Descriptive statistics and models of window size trends. Values correspond to dimensional characterization of parts with different unit cell scale, fabricated from a fully-loaded print, as shown in Fig. 8.

Model	Median A_w/A_{design}	Result
1.86 mm window	0.935	$A_w/A_{design} = 0.008 R + 0.854$
1.3 mm window	0.928	$A_w/A_{design} = 0.016 R + 0.772$
0.74 mm window	0.906	$A_w/A_{design} = 0.039 R + 0.523$

to the results observed here; differences in trends are readily observable in the model coefficients and representation of the data. However, the apparent discrepancy does not imply disagreement. In addition to differences in part geometry, we examine features at smaller length scales than their studies, which may contribute to the observed differences. In fact, our results suggest that the overcure effect may be more pronounced as the unit cell length is scaled down. We hypothesize that as the unit cell length scales decrease, the proximity of the polymerized regions contribute to radical and thermal gradients, which, when coupled with aberrant light scattering, increase the likelihood of local curing effects. The linear models were not compared for statistical significance due to differences in sample size between lattice dimensions and the influences of the fabrication process, where platform position and post-processing steps could impact the final part dimensions. Regardless, these considerations warranted further investigation into the influence of part position and reproducibility between parts, under conditions as close to ideal as realizable.

To examine dimensional reproducibility as well as the possible effects of part positioning and the number of parts printed simultaneously, a series of prints was performed to compare a single central part, a central part with multiple parts loaded on the print area, and an off-center part with multiple parts loaded on the print area. A single off-center part was not considered in this case because such a configuration is less representative of typical processing conditions or manufacturing conditions where multiple parts would be printed

simultaneously. These parts were printed in UMA 90 Black for improved dimensional fidelity, as recommended by the manufacturer. The results for each configuration are shown in Fig. 9. Across all configurations, window sizes consistently increase with increasing radial distance by approximately 12%/mm, beginning at approximately 95% of the designed window area at the center of the part and increasing to approximately 110% of the designed window area at 14 mm from the center of the part. Statistically, the trend of window size with radial distance is not statistically significantly different between the three part configurations ($p = 0.332$), while there is a statistically significant difference in the mean values and model intercepts ($p < 0.001$), shown in Table 2.

Care was taken to use the same hardware within a similar timeframe for production of all parts, and to make post-processing treatments effectively identical between parts. The observed position dependence for part dimensions is consistent with previous reports for a similar printing system and material [49]. A notable trend difference between these parts and the UMA 90 Cyan parts is the deviation from curing. Both colors show a window size increase from the center of the part to the periphery, but the darker colored resin starts at a closer value to the design value. The darker resin's normalized window area exceeds unity at the periphery, instead of approaching the design value as shown for UMA 90 Cyan. This is an important consideration in choosing the color of the resin to be printed. Since all colors in the UMA 90 family are printed using the same settings, these results suggest that pigment additives can have significant effects on absorption properties of the resin and influence part dimensions for small feature sizes. Despite the statistical significance between window area at the core of the parts based on position and part loading, this effect is small compared to the variation from part center to periphery between all parts. In the context of dimensional accuracy to the as-designed part, the difference in center dimension was not considered physically significant compared to phenomena like the radial trend, which is nearly an order of magnitude larger. The material impact of this phenomena may be negligible where lattices are used for maintaining structural integrity, but it can lead to undesirable results in applications that depend on void space, rather than part volume.

4.4. Dimensional compensation

An iterative process was employed to adjust the CAD model in response to variation in measured window area occurring from the radial overcure. After imaging the lattice parts (the first, corresponding to the 1 Part, Center results in Fig. 9, is shown in Fig. 10A), the normalized deviation in approximate strut radius ($\Delta r_{\text{approx}}/r$) was

Table 2

Models and ANCOVA results of platform loading and positioning. Values correspond to dimensional characterization of the three lattice configurations as shown in Fig. 9.

Model	Result
1 Part, Center	$A_w/A_{\text{design}} = 0.0120 R + 0.961$
3 Parts, Center	$A_w/A_{\text{design}} = 0.0125 R + 0.967$
3 Parts, Off-center	$A_w/A_{\text{design}} = 0.0122 R + 0.942$
ANCOVA, Slope Heterogeneity	$P = 0.332$
ANCOVA, Mean Equivalence	$P < 0.001$

modeled as a function of radial distance. This model was used to produce the mathematical strut radius compensation shown in Table 3. The primary objective of compensation was to reduce the trend with radial distance, so offset terms were neglected. Reconstructed micrographs and normalized strut radius deviation of the as-printed parts at each iteration of compensation are shown in Fig. 10. With the first iteration (Fig. 10B), compensation overshoots the target strut radius and produces a part where r_{approx} increases by approximately 1.1%/mm. After repeating the process for a second iteration (Fig. 10C), r_{approx} decreases by approximately 0.2%/mm. This trend is statistically significant, but it demonstrates an approximately 85% reduction in comparison to the original printed parts. Furthermore, these results illustrate the ability of this approach to correct for both increases and decreases in strut thickness with radial dimension, a phenomenon which may vary between resin chemistries, as shown in the differences between UMA Black here and UMA Cyan in Section 4.3.

Specific to the 2D approach utilized in this case, the compensation method is effective in decreasing the slope of window area with radial distance, but the final results also suggest a degree of variability comparable to the original part. This may arise from multiple processing steps, but the largest contributor is likely the process of approximating the strut radius. In this method, the approximate radius is a virtual value at the center of each window, which is then mapped to the actual strut points. As a result, the struts on all sides of the window average to the center virtual strut radius, but this may cause deviations in the final printed result. An improvement over this approach would be to use a 3D imaging technique like X-ray computed tomography (CT). In the 3D case, the same compensation approach could be applied by modeling the deviation as a function of radial distance and position along the print axis. The 3D imaging modality naturally yields a more direct measurement of strut radius, rather than an approximation, which can account for curing effects that occur in the Z direction and reduce deviations that arise from applying the approximation to off-axis struts. Furthermore, the use of CT-based models for strut deviation can extend the application

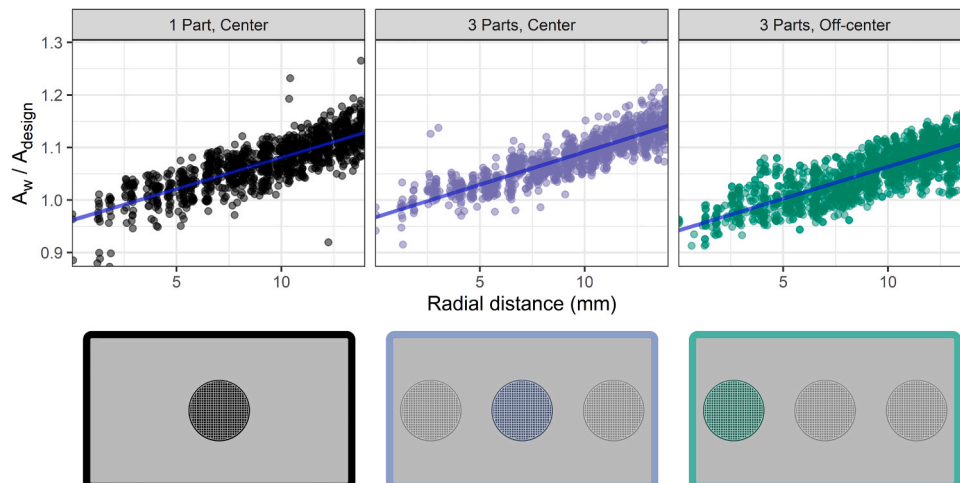


Fig. 9. Dimensional characterization of the three lattice configurations ($n = 3$ replicates per configuration).

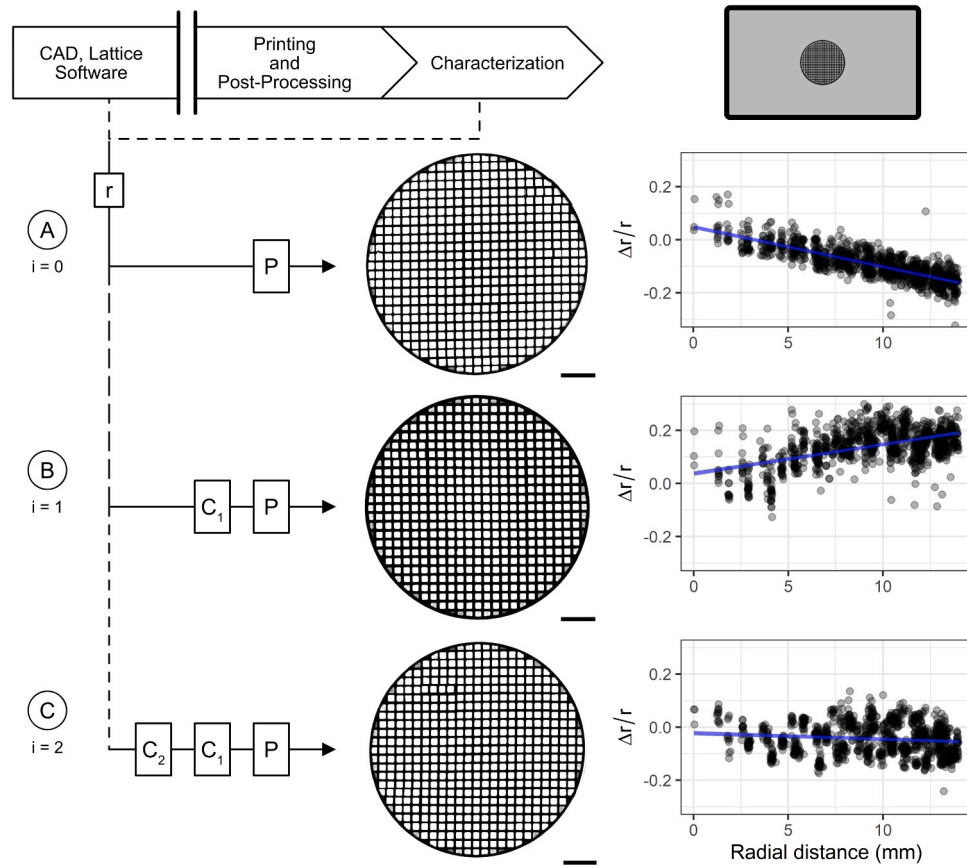


Fig. 10. Compensation schemes and strut radius deviation from each iteration of curing compensation ($n = 3$ replicates per iteration). (A) First iteration, no compensation. (B) Second iteration, single-stage compensation. (C) Third iteration, two-stage compensation. Scale bar is 5 mm.

Table 3

Compensation functions and strut radius models used in curing compensation. Values correspond to compensation schemes from each iteration of curing compensation as shown in Fig. 10.

Model	Input strut radius function	Result
1 Part, Center	$r_0 = 0.183$	$\Delta r_{\text{approx}}/r_0 = -0.015 R + 0.048$
Compensation 1	$r_1 = 0.183 \left(\frac{1}{1 - 0.015R} \right)$	$\Delta r_{\text{approx}}/r_0 = 0.011 R + 0.037$
Compensation 2	$r_2 = 0.183 \left(\frac{1}{1 - 0.015R} \right) \left(\frac{1}{1 + 0.011R} \right)$	$\Delta r_{\text{approx}}/r_0 = -0.002 R - 0.023$

of such a compensation approach to other unit cell geometries, where most struts do not lie in the same plane for printing and imaging.

Compensation methods have been successfully implemented for other additive processes with traditional compensation techniques and models of part deviation [22,50]. Because of various process influences, the general approaches for compensation and closed-loop control can be widely applicable, but parameters may need to be specified on an individual basis, taking into account fundamental process models, hardware variability, part orientation, and post-processing steps [48,49,51]. However, the requisite information may not be available to an end-user, especially when proprietary tools and materials are involved. For these reasons, we elected to adapt process engineering principles in this two-stage design. The approach here effectively defines one compensator for the process (which includes digital aspects like slicing and physical aspects like printing and post-processing) and one compensator to account for unknown effects that arise from the adjustment to the CAD dimensions (shown in the block diagram in Fig. 10). As a result, this approach can be better-suited for low-volume production or prototyping. These are situations where a large number of iterations may not be practical, or a computational model is not justifiable or available. One

example is the use of different resins with a common part geometry. In this case, resin conditions such as color, mechanical properties, or the necessity of thermal curing could influence the final part's dimensions and sensitivity to input changes. The compensation parameters and total number of iterations required may vary between resin chemistries, but a naïve compensation approach such as this one would be well-suited for this situation because it does not need to take resin properties into account and it does not assume information about the process. Therefore, while the compensation values may not be transferrable between materials, geometries, printers, or processing steps, the general approach remains broadly applicable.

One limitation of this approach is that it assumes a linear trend based on the initial part results, which may not hold for all configurations of material, geometry, and dimension. Furthermore, if a large number of iterations is required to achieve a sufficient level of compensation, the resulting polynomial compensation model may be impractical. Finally, process capability and print resolution are also critical factors for effective compensation. If the required change is smaller than the printer resolution or the process variability exceeds the tolerance set for a given application, compensation will ultimately prove ineffective. Similarly,

the number of required iterations is highly dependent on the application's dimensional tolerances. Factors such as part location or tool performance can also impact the number of iterations required to achieve an appropriate level of compensation, which will vary depending on the application design tolerances. McGregor et al. presents a thorough statistical analysis of variability in the class of printers used here [44,49]. Despite this, the general approach remains broadly applicable and can be augmented as more information becomes available. Correcting for digital and physical processing is challenging because of the numerous inputs and effects that contribute to the final parts, but with advances in numerical methods for print process modeling, effective feedforward and closed-loop control schemes will be able to maximize the fidelity of as-printed products with respect to their as-designed counterparts.

5. Conclusion

Lattices are a unique opportunity in additive manufacturing, but the separation between digital and physical processing creates additional barriers to access, design exploration, and ultimately utility. Moreover, controlling feature sizes and achieving small feature dimensions are critical for new applications. Here we present a full pipeline for process-oriented open lattices, including design methodology, printability assessment, dimensional characterization, and a low-volume compensation strategy. The open lattice framework is presented for traditional node-strut lattices including cubic, Kelvin, BCCxy, RD, and WP unit cells. Its robustness is further demonstrated with a conceptual representation of two curved-beam lattices, which may have implications for flexible lattice applications. The results of the printability assays show that for this manufacturing system, lattices can be scaled down to the limit of printer resolution, which may be below manufacturer-recommended dimensions. Furthermore, we observe that porosity and strut radius can be used as partial predictors of print success for bending-dominated lattice structures. The same assay can easily be applied to a range of lattice structures and other manufacturing systems as well, which is significant for determining the possible extent of lattice scaling, increasing design throughput, and providing process-informed design rules for lattice generation software. With respect to other physical processing stages, we observe increased feature size deviations as lattice geometries decrease in length scale, and we observe that resin chemistry and part location during the print effect the trends in part deviation for the vat photopolymerization process used. To compensate for process-influenced deviations, we implement a simplified iterative feedforward scheme for functionally grading the strut dimension along the radial direction. Compensated designs significantly reduce the radial dependence of as-printed lattice window area after just two iterations. Based on the finding that radial dependence may not vary significantly with part location, this compensation scheme could be successfully applied a suite of parts in the same batch, with only moderate tuning.

In summation, this work is one of the first comprehensive descriptions of open lattice generation and processing and the results presented here are promising for the future of fully open lattice structures as a complementary design approach to current offerings. The design strategy is computationally minimal to accommodate a range of systems and it can be further optimized when symmetrical features are available. The self-supporting, open structure means less material waste in the printing and post-processing steps. Future work is necessary to examine the impact of the open skin and trimmed beams with respect to structurally optimized geometries and non-structural applications. With further investigation of the printable range of these lattices across other additive manufacturing systems, in conjunction with advanced process models, there is great potential for these structures to be utilized in a broad range of contexts and realize the additive manufacturing vision of rapid, on-demand parts specific to any use case.

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CRediT authorship contribution statement

Ian R. Woodward: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Catherine A Fromen:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.addma.2021.102386](https://doi.org/10.1016/j.addma.2021.102386).

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